

devt

Package in package 'Model'

devt

Version 0.4.0 Phase 1.0 Proposed

jkl created on 06.03.2026. Last modified 02.10.2026

wmdr2-devt diagram

Class diagram in package 'devt'

WMDR2 v0.4.x

v0.4.0

- Color type boundaries to emphasize alignment with wmdr2 1.99.dev1 and distinguish remaining additions. Red boundaries denote wmdr2 1.99.dev1 types, * denote attributes present in 1.99.dev1. Types with blue shading denote re-usable types. NB: the Contact type is provided by OGC API Records directly and is not a WMDR2 addition.

- add Contact.roles[0..1] to further align with 1.99.dev1. Propose specific role codes to be amended to the <http://codes.wmo.int/wis/contact-role>.

Differences wmdr2 1.99.dev1 vs wmdr2-devt v0.4.0

Path | 1.99.dev1 | v0.4.0

Facility.links | 1..* | 0..*

Facility.territories | 0..* | at least one for fixed facility

Observations.observedGeometry | 0..1 | 1

Instrument.manufacturer | 0..1 | 1

Instrument.model | 0..1 | 1

wmdr2-devt

Version 0.4.0

jkl created on 06.03.2026. Last modified 02.10.2026

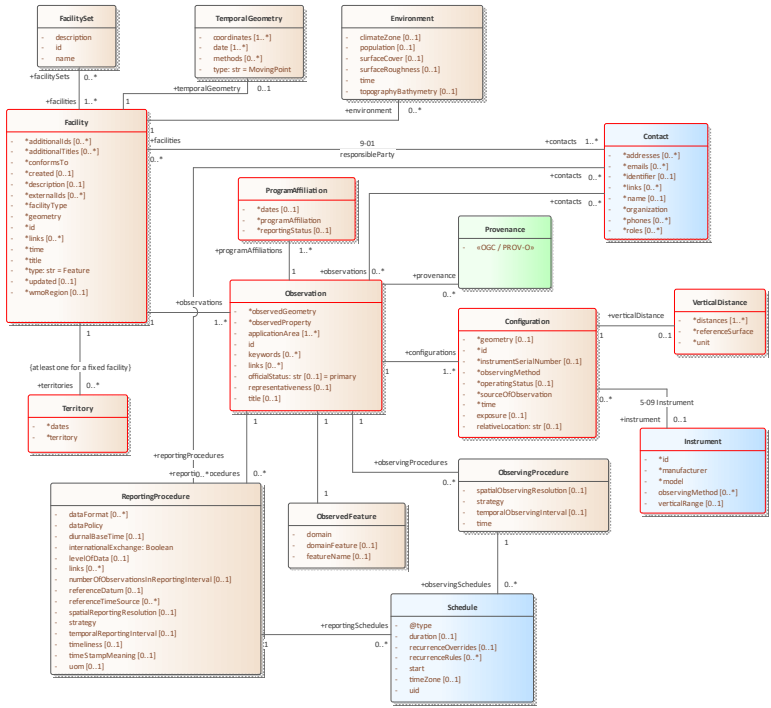


Figure 1: wmdr2-devt

Configuration

Class in package 'devt'

WMDR2 core JSON propoerty: configurations.
Describes the historical configuration of the observations during a specified time period.

Configuration
Version 1.0 Phase 1.0 Proposed
jkl created on 26.06.2026. Last modified 21.09.2026

ATTRIBUTES
*geometry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) geometry is the OGC Records name for geolocation.

ATTRIBUTES	
	For stationary facilities, this represents the geolocation of the sensor / inlet, and should be documented. If not specified, the geometry (and temporalGeometry) of the facility is assumed. For mobile facilities, this should be left blank. [Is static True. Containment is Not Specified.]
 *id : Private	[Is static True. Containment is Not Specified.]
 *instrumentSerialNumber : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 *observingMethod : Private from (new) unified code list of observing methods, based on http://codes.wmo.int/wmdr/ObservingMethodAtmosphere http://codes.wmo.int/wmdr/ObservingMethodTerrestrial http://codes.wmo.int/wmdr/ObservingMethodEarth	[Is static True. Containment is Not Specified.]
 *operatingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) values refer to http://codes.wmo.int/wmdr/InstrumentOperatingStatus	[Is static True. Containment is Not Specified.]
 *sourceOfObservation : Private Value from http://codes.wmo.int/wmdr/SourceOfObservation. If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null.	[Is static True. Containment is Not Specified.]
 *time : Private conformant to OGC time: [beginDate, endDate]	[Is static True. Containment is Not Specified.]
 exposure : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/Exposure	[Is static True. Containment is Not Specified.]
 relativeLocation : str Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Free text, documenting the relative location of instrument / sensor / inlet /observer with respect to the geometry (and temporalGeometry) of the facility.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	

ASSOCIATIONS	
✍ Association (direction: Unspecified) 5-09 Instrument	
Source: Public (Class) Configuration Cardinality: [0..*]	Target: Public instrument (Class) Instrument Cardinality: [0..1]
✍ Association (direction: Unspecified)	
Source: Public (Class) Configuration Cardinality: [1]	Target: Public verticalDistance (Class) VerticalDistance Cardinality: [0..1]
✍ Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public configurations (Class) Configuration Cardinality: [1..*]

Contact

Class in package 'devt'

WMDR2 core JSON class: contact.

Contact associations carry one or more contextual **roles**. In JSON, the relationship is serialized as a `contactAssignments` object containing the contact reference and role(s).

Contact

Version 1.0 Phase 1.0 Proposed

jkl created on 11.03.2026. Last modified 25.09.2026

ATTRIBUTES	
💡 *addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Postal addresses.	[Is static True. Containment is Not Specified.]
💡 *emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Email contacts.	[Is static True. Containment is Not Specified.]
💡 *identifier : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Optional stable contact identifier.	[Is static True. Containment is Not Specified.]
💡 *links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	

ATTRIBUTES	
Online resources associated with the contact. [Is static True. Containment is Not Specified.]	
*name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard) [Is static True. Containment is Not Specified.]	
*organization : Private Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss [Is static True. Containment is Not Specified.]	
phones : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) Phone contacts. [Is static True. Containment is Not Specified.]	
roles : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) roles[] follows http://codes.wmo.int/wis/contact-role. The development terms supervisor, custodian, pointOfContact, and principallInvestigator are proposed additions to that vocabulary and are used provisionally by wmdr2-devt pending amendment of the WCMP codelist. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) 9-01 responsibleParty Source: Public facilities (Class) Facility Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [1..*]	
Association (direction: Unspecified) Source: Public observations (Class) Observation Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*] Target: Public contacts (Class) Contact Cardinality: [0..*]	

Environment

Class in package 'devt'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

Environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 02.06.2026. Last modified 25.06.2026

ATTRIBUTES	
climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility [Is static True. Containment is Not Specified.]	
surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
time : Private conformant to OGC time: [beginDate, endDate] [Is static True. Containment is Not Specified.]	
topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public environment (Class) Environment Cardinality: [0..*]	

Facility

Class in package 'devt'

Facility properties payload of the WMDR2 full record. JSON serialization is split between GeoJSON root and properties.

ATTRIBUTES	
additionalIds : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) Additional WSLs for the facility. [Is static True. Containment is Not Specified.]	
additionalTitles : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) Additional names for the facility, i.e., "aliases" that may be in use, maybe by specific programs. [Is static True. Containment is Not Specified.]	
*conformsTo : Private [Is static True. Containment is Not Specified.]	
*created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata. [Is static True. Containment is Not Specified.]	
*description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description. [Is static True. Containment is Not Specified.]	
externalIds : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) Generic OGC Records metadata. External identifiers such as used internally by an organization's data management system; also used to document programSpecificFacilityId. In that case, the "scheme" should be a recognized program, e.g., "externalIds": [{ "scheme": "https://codes.wmo.int/wmdr/ProgramAffiliation/GAWglobal", "value": "JFJ" }, { "scheme": "https://codes.wmo.int/wmdr/ProgramAffiliation/NDACC", "value": "Jungfrauoch, Switzerland" }] Note: Any programSpecificFacilityTitle should be listed under additionalTitles. [Is static True. Containment is Not Specified.]	
*facilityType : Private Alias: 3-04 facilityType	

ATTRIBUTES	
WMDR facility type code value. [Is static True. Containment is Not Specified.]	
*geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation [Is static True. Containment is Not Specified.]	
*id : Private Alias: 3-06 (primary) WSI The primary WIGOS Station Identifier [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..], Allow duplicates: 0, Is ordered: False) Related links. [Is static True. Containment is Not Specified.]	
*time : Private Alias: [dateEstablished, dateClosed] mandatory element in OGC Records API Part 1, an interval. Used to declare [dateEstablished, dateClosed] [Is static True. Containment is Not Specified.]	
*title : Private Alias: 3-03 facilityName properties.title = 3-03 facilityName [Is static True. Containment is Not Specified.]	
*type : str Private = Feature Facility properties type; const facility. [Is static True. Containment is Not Specified.]	
*updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata. [Is static True. Containment is Not Specified.]	
*wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
Association (direction: Unspecified) 9-01 responsibleParty	
Source: Public facilities (Class) Facility Cardinality: [0..*]	Target: Public contacts (Class) Contact Cardinality: [1..*]
Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public observations (Class) Observation Cardinality: [1..*]
Association (direction: Unspecified)	
Source: Public facilities (Class) Facility Cardinality: [1..*]	Target: Public facilitySets (Class) FacilitySet Cardinality: [0..*]
Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public environment (Class) Environment Cardinality: [0..*]
Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public temporalGeometry (Class) TemporalGeometry Cardinality: [0..1]
Association (direction: Unspecified)	
Constraints: at least one for a fixed facility : Invariant	
Source: Public (Class) Facility Cardinality: [1]	Target: Public territories (Class) Territory Cardinality: [0..*]

FacilitySet

Class in package 'devt'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

FacilitySet

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
description : Private	[Is static True. Containment is Not Specified.]

ATTRIBUTES	
 id : Private [Is static True. Containment is Not Specified.]	
 name : Private Alias: title [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public facilities (Class) Facility Cardinality: [1..*] Target: Public facilitySets (Class) FacilitySet Cardinality: [0..*]	

Instrument

Class in package 'devt'

Describes instrument capabilities, not a specific instrument instance.
Instrument are re-usable between configurations and even facilities.

Instrument

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
 *id : Private [Is static True. Containment is Not Specified.]	
 *manufacturer : Private Alias: 5-09 manufacturer [Is static True. Containment is Not Specified.]	
 *model : Private Alias: 5-09 model [Is static True. Containment is Not Specified.]	
 observingMethod : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observing method(s) an instrument may expose. This is meant to support users in selecting the right instrument from a catalogue. [Is static True. Containment is Not Specified.]	
 verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.	

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 5-09 Instrument	
Source: Public (Class) Configuration Cardinality: [0..*]	Target: Public instrument (Class) Instrument Cardinality: [0..1]

Observation

Class in package 'devt'

WMDR2 core JSON class: observations.
Describes a series of observations of a specific observedProperty with its observedGeometry.

Observation

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
 *observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry NB: WMDR2 1.99.dev1 includes this as observedGeometry[0..1], so optional! [Is static True. Containment is Not Specified.]	
 *observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2 . NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll - observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text. - matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ... These elements will be supported by code lists from codes.wmo.int/wmdr2 . Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
applicationArea : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Value(s) from http://codes.wmo.int/wmdr/ApplicationArea [Is static True. Containment is Not Specified.]	
id : Private Observation identifier, e.g. "12006-point". [Is static True. Containment is Not Specified.]	
keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords. [Is static True. Containment is Not Specified.]	
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links. [Is static True. Containment is Not Specified.]	
officialStatus : str Private = primary Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/ObservationStatus Defaults to "primary" if not provided. [Is static True. Containment is Not Specified.]	
representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness [Is static True. Containment is Not Specified.]	
title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional label. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public observingProcedures (Class) ObservingProcedure Cardinality: [0..*]	
Association (direction: Unspecified)	

ASSOCIATIONS	
Source: Public (Class) Observation Cardinality: [1]	Target: Public programAffiliations (Class) ProgramAffiliation Cardinality: [1..*]
✍ Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public configurations (Class) Configuration Cardinality: [1..*]
✍ Association (direction: Unspecified)	
Source: Public observations (Class) Observation Cardinality: [0..*]	Target: Public contacts (Class) Contact Cardinality: [0..*]
✍ Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public (Class) ObservedFeature Cardinality: [1]
✍ Association (direction: Unspecified)	
Source: Public (Class) Observation	Target: Public provenance (Class) Provenance Cardinality: [0..*]
✍ Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]
✍ Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public observations (Class) Observation Cardinality: [1..*]

ObservedFeature

Class in package 'devt'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

ObservedFeature

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES
🔒 domain : Private

ATTRIBUTES	
The largest scale in WIGOS, cf. http://codes.wmo.int/wmdr/Domain [Is static True. Containment is Not Specified.]	
domainFeature : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation. Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like - river basin - lake - river - catchment - canopy - epilimnion - hypolimnion - troposphere - tropopause - stratosphere - thermocline (useful as a general term, but needs "domain") - boundary layer (useful as a general term, but needs "domain") - ... [Is static True. Containment is Not Specified.]	
featureName : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin" [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public (Class) ObservedFeature Cardinality: [1]	

ObservingProcedure

Class in package 'devt'

When are observations made? Describes an observingStrategy, and depending on it, a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".
Implement as array of objects

ObservingProcedure

Version 1.0 Phase 1.0 Proposed

jkl created on 11.03.2026. Last modified 25.09.2026

CONSTRAINTS
Invariant. {temporalObservingInterval is specified OR at least one observingSchedule is specified} [Approved, Weight is 0.]

ATTRIBUTES
spatialObservingResolution : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either a distance or a grid / pixel -dimension. [Is static True. Containment is Not Specified.]
strategy : Private Values from http://codes.wmo.int/wmdr/SamplingStrategy [Is static True. Containment is Not Specified.]
temporalObservingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) ISO-8601 duration, e.g. - "PT10M" for 10' intervals - "PT1H" for 1-hour intervals - "P1D" for daily intervals [Is static True. Containment is Not Specified.]
time : Private conformant to OGC time: [beginDate, endDate] [Is static True. Containment is Not Specified.]

ASSOCIATIONS
Association (direction: Unspecified) Source: Public (Class) ObservingProcedure Cardinality: [1] Target: Public observingSchedules (Class) Schedule Cardinality: [0..*]
Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public observingProcedures (Class) ObservingProcedure Cardinality: [0..*]

ProgramAffiliation

Class in package 'dev'

ATTRIBUTES	
 *dates : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]	
 *programAffiliation : Private [Is static True. Containment is Not Specified.]	
 *reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Observation Cardinality: [1] Target: Public programAffiliations (Class) ProgramAffiliation Cardinality: [1..*]	

Provenance

Class in package 'devt'

[TODO] Provenance is supplied by the OGC provenance building block

- aggregationInterval [0..1]: Describes the smallest interval over which raw/ level 0 observations are aggregated and stored. Express as ISO-8601 duration.

ATTRIBUTES	
 «OGC / PROV-O» : Private [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Observation Target: Public provenance (Class) Provenance Cardinality: [0..*]	

ReportingProcedure

Class in package 'devt'

ReportingProcedure

Version 1.0 Phase 1.0 Proposed

jkl created on 23.06.2026. Last modified 25.09.2026

CONSTRAINTS
Invariant. {temporalReportingInterval is specified OR at least one reportingSchedule is specified} [Approved, Weight is 0.]
ATTRIBUTES
dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat. [Is static True. Containment is Not Specified.]
dataPolicy : Private Value from http://codes.wmo.int/wmdr/DataPolicy. JSON Concept or explicit null where permitted. [Is static True. Containment is Not Specified.]
diurnalBaseTime : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]
internationalExchange : Boolean Private Declaration if data are intended for international exchange or not. [Is static True. Containment is Not Specified.]
levelOfData : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/LevelOfData. [Is static True. Containment is Not Specified.]
links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval. [Is static True. Containment is Not Specified.]
numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]

ATTRIBUTES	
referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem. [Is static True. Containment is Not Specified.]	
referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime. [Is static True. Containment is Not Specified.]	
spatialReportingResolution : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either a distance or a grid / pixel -dimension. [Is static True. Containment is Not Specified.]	
strategy : Private [Is static True. Containment is Not Specified.]	
temporalReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) ISO-8601 duration, e.g. "PT1H" [Is static True. Containment is Not Specified.]	
timeliness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Expected timeliness of reporting, an object with value and uom (default: seconds) [Is static True. Containment is Not Specified.]	
timeStampMeaning : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/TimeStampMeaning. [Is static True. Containment is Not Specified.]	
uom : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Unit of reported values of the observedProperty. Values from http://codes.wmo.int/wmdr/unit. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified)	

ASSOCIATIONS	
Source: Public (Class) ReportingProcedure Cardinality: [1]	Target: Public reportingSchedules (Class) Schedule Cardinality: [0..*]
✓ Association (direction: Unspecified)	
Source: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]	Target: Public contacts (Class) Contact Cardinality: [0..*]
✓ Association (direction: Unspecified)	
Source: Public (Class) Observation Cardinality: [1]	Target: Public reportingProcedures (Class) ReportingProcedure Cardinality: [0..*]

Schedule

Class in package 'devt'

Reusable JSCalendar Event schedule to document temporal coverage and exceptions.

Schedule

Version 1.0 Phase 1.0 Proposed

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ATTRIBUTES	
🔒 @type : Private JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
🔒 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]
🔒 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid.	[Is static True. Containment is Not Specified.]
🔒 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. Use start, duration and other elements to evaluate constraints if needed.	[Is static True. Containment is Not Specified.]
🔒 start : Private	

ATTRIBUTES	
Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	

ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) ObservingProcedure Cardinality: [1] Target: Public observingSchedules (Class) Schedule Cardinality: [0..*]	
Association (direction: Unspecified) Source: Public (Class) ReportingProcedure Cardinality: [1] Target: Public reportingSchedules (Class) Schedule Cardinality: [0..*]	

TemporalGeometry

Class in package 'devt'

TemporalGeometry

Version 1.0 Phase 1.0 Proposed

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Alias temporalGeometry

ATTRIBUTES	
coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
date : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	

ATTRIBUTES	
methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
type : str Private = MovingPoint [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public temporalGeometry (Class) TemporalGeometry Cardinality: [0..1]	

Territory

Class in package 'devt'

WMDS reference: 3-02
mandatory for fixed facilities, optional for mobile facilities.
JSON serializes the temporal extent as official dates.

Territory

Version 1.0 Phase 1.0 Proposed

jkl created on 23.06.2026. Last modified 25.09.2026

ATTRIBUTES	
*dates : Private conformant to OGC time: [beginDate, endDate] [Is static True. Containment is Not Specified.]	
*territory : Private Alias: temporal Territory Values from http://codes.wmo.int/wmdr/TerritoryName [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
Association (direction: Unspecified) Constraints: at least one for a fixed facility : Invariant Source: Public (Class) Facility Cardinality: [1] Target: Public territories (Class) Territory Cardinality: [0..*]	

ASSOCIATIONS

VerticalDistance

Class in package 'devt'

VerticalDistance

Version 1.0 Phase 1.0 Proposed

jkl created on 26.06.2026. Last modified 21.09.2026

ATTRIBUTES
distances : Private Multiplicity: ([1..], Allow duplicates: 0, Is ordered: False) [Is static True. Containment is Not Specified.]
*referenceSurface : Private [Is static True. Containment is Not Specified.]
*unit : Private [Is static True. Containment is Not Specified.]

ASSOCIATIONS
Association (direction: Unspecified) Source: Public (Class) Configuration Cardinality: [1] Target: Public verticalDistance (Class) VerticalDistance Cardinality: [0..1]